

Automation Exercise – Capstone Project

Overview

1. Project Introduction

This project is an end-to-end automation testing framework built using the Automation Exercise website. It focuses on validating core e-commerce functionalities through UI and API automation testing.

The framework is designed using the Page Object Model (POM) with Playwright and JavaScript to ensure scalability, maintainability, and readability.

2. Project Objective

The main goal of this project is to build an end-to-end automation testing framework for an e-commerce application using the Automation Exercise website.

It validates real user workflows using both UI and API automation to ensure functionality, reliability, and data accuracy.

Tools Used: Playwright, JavaScript (Node.js), Page Object Model (POM), API Testing (Playwright Request)

5. Test Coverage (8 Services)

Each module is structured in a tabular format with Service Name, Number of Test Cases, Priority, and What is Tested.

Service Module	Test Cases	Priority	What is Tested
Authentication Module	15	Critical	Signup, login, logout, forgot password, and input validation for user authentication flow using UI automation
Product Module	20	High	Product listing, search functionality, product details page, category filtering, and product information validation
Cart Module	15	Critical	Add to cart, remove items, update quantity, cart total calculation, and empty cart validation
Checkout & Order Module	20	Critical	Checkout flow, address confirmation, payment process, order placement, and order summary validation

Service Module	Test Cases	Priority	What is Tested
User Profile Module	10	Medium	Profile update, email update, password change, and verification of updated user data
Shipping Address Module	10	Medium	Add, edit, delete shipping address, and validation of required address fields
Customer Support Module	10	Low	Contact form submission, form validation, and successful message submission confirmation
API Testing Module	20	High	GET and POST APIs for products, login, signup, cart operations, and response validation

6. Total Test Case Summary Customer Support Module

- **Test Cases:** 10
- **What is Tested:** Contact form submission, form validation, and successful message submission confirmation.

8. API Testing Module

- **Test Cases:** 20
- **What is Tested:** GET and POST APIs for products, login, signup, cart operations, and response validation.

7. Total Test Case Summary Customer Support Module (10 TC)

Tests contact form submission, validation checks, and message confirmation.

8. API Testing Module (20 TC)

Validates backend APIs including GET/POST requests for products, login, signup, and cart operations.

7. Total Test Case Summary Customer Support Module (10 Test Cases)

Goal: Ensure proper communication channel between user and support team.

What is tested:

- Contact form submission
- Form validation checks
- Successful message submission
- Email confirmation response

8. API Testing Module (20 Test Cases)

Goal: Validate backend API functionality and response correctness.

What is tested:

- GET product API responses
 - POST login API validation
 - POST signup API
 - Cart-related API operations
 - Response status and data validation
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7. Total Test Case Summary

Module	Test Cases
Authentication	15
Product Module	20
Cart Module	15
Checkout & Order	20
User Profile	10
Shipping Address	10
Customer Support	10
API Testing	20
Total	120

9. Key Features

- Page Object Model (POM) implementation
 - UI + API hybrid automation
 - Data-driven testing approach
 - Parallel test execution support
 - Scalable test framework design
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10. Execution Commands

```
npm install
npx playwright install
```

```
npx playwright test  
npx playwright show-report
```

11. Conclusion

This automation project demonstrates a complete real-world testing framework covering UI and API testing. It ensures end-to-end validation of an e-commerce application and follows industry-standard automation practices using Playwright and POM architecture.